

KON 313

Feedback Control Systems

Steady-State Errors (e_{ss})

Dr. Aysa Jafari Farmand

aysa.jafari@bilgi.edu.tr

Contents

- Introduction
- Steady-State Errors for Unity Feedback Systems
- Static Error Constants and System Type
- Steady-State Error Specifications
- Steady-State Error for Nonunity Feedback Systems
- Steady-State Error due to Disturbances

Introduction

Recall. Control systems analysis and design focus on three specifications: (1) transient response, (2) stability, and (3) steady-state errors.

Now we are ready to examine **steady-state** errors.

We define the errors and derive methods of controlling them.

As we progress, we find that control system design entails **trade-offs** between desired transient response, steady-state error, and the requirement that the system be stable.

Our steady state analysis are limited to **stable** systems, where the natural response approaches zero as $t \rightarrow \infty$. Unstable systems represent loss of control in the steady state and are not acceptable for use at all.

The steady-state error calculations can contain errors for an unstable system. Thus, we must check the system for stability while performing steady-state error analysis and design. **Here, we assume that all the systems are stable.**

Introduction: Definition

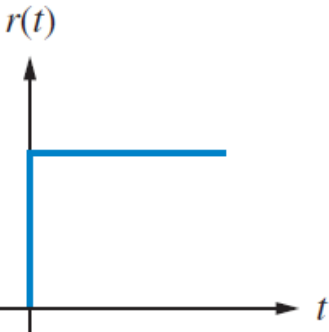
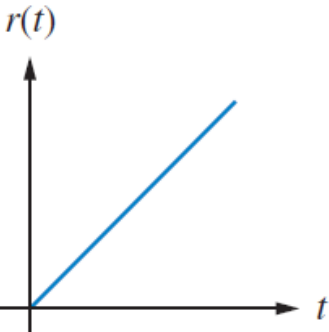
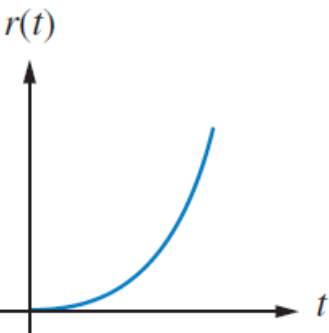
Steady-state error, e_{ss} , is the difference between the input and the output for a prescribed test input as $t \rightarrow \infty$. **Test inputs** used for steady-state error analysis and design are as follows.

In order to explain how these test signals are used, let us assume a **position control system**, where the output position follows the input commanded position.

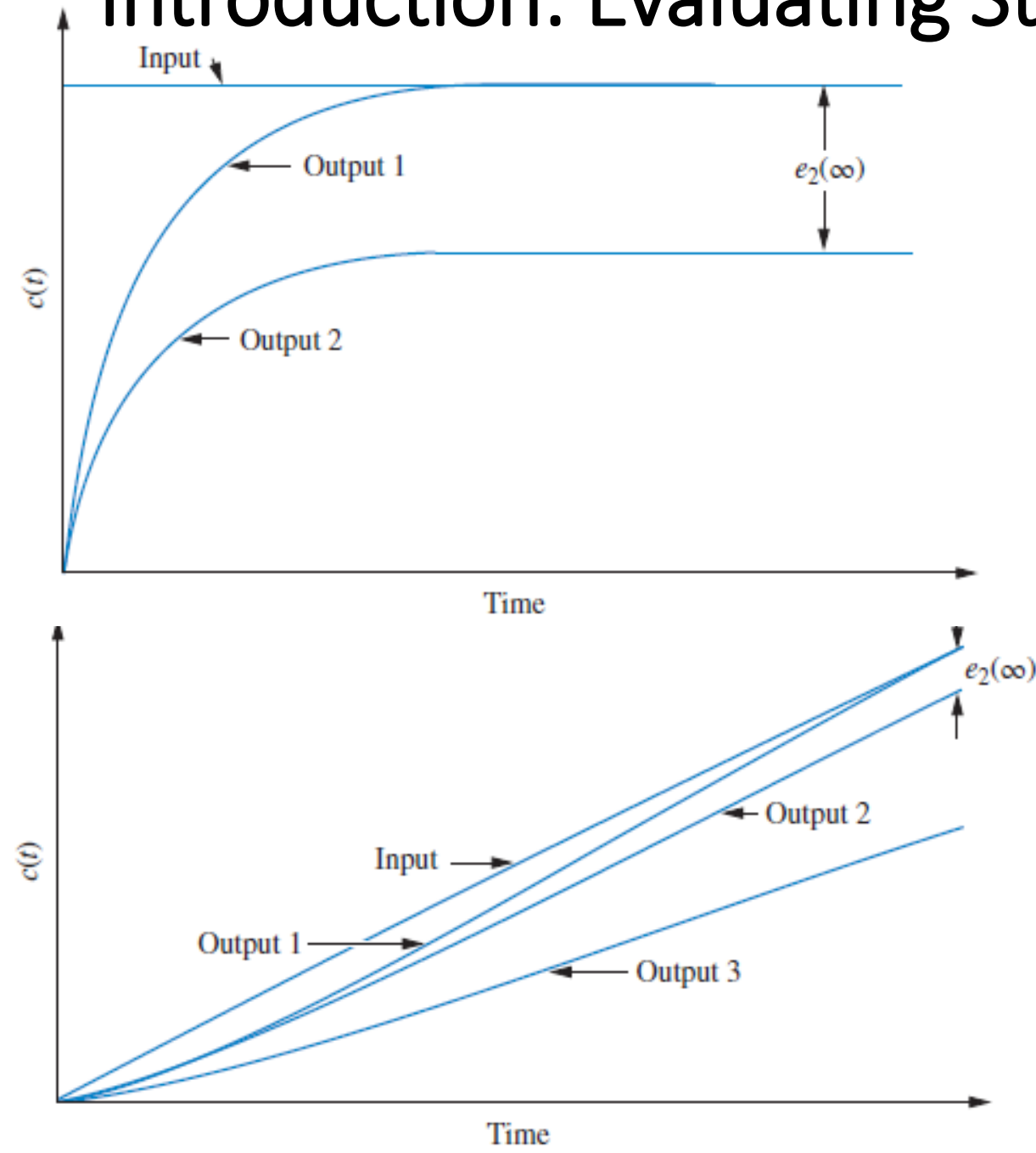
- **Step inputs** represent **constant position** and thus are useful in determining the ability of the control system to position itself with respect to a stationary target.
- **Ramp inputs** represent **constant-velocity** inputs to a position control system by their linearly increasing amplitude. These waveforms can be used to test a system's ability to follow a linearly increasing input or, equivalently, to track a constant velocity target.
- **Parabolas**, whose second derivatives are constant, represent **constant acceleration** inputs to position control systems and can be used to represent accelerating targets.

Introduction: Definition

The waveforms for evaluating steady-state error

Waveform	Name	Physical interpretation	Time function	Laplace transform
 A graph of a step function $r(t)$ versus time t. The vertical axis is labeled $r(t)$ and the horizontal axis is labeled t. The function is zero for $t < 0$ and jumps to a constant positive value for $t \geq 0$.	Step	Constant position	1	$\frac{1}{s}$
 A graph of a ramp function $r(t)$ versus time t. The vertical axis is labeled $r(t)$ and the horizontal axis is labeled t. The function is zero for $t < 0$ and increases linearly with a constant positive slope for $t \geq 0$.	Ramp	Constant velocity	t	$\frac{1}{s^2}$
 A graph of a parabolic function $r(t)$ versus time t. The vertical axis is labeled $r(t)$ and the horizontal axis is labeled t. The function is zero for $t < 0$ and increases quadratically with a constant positive slope for $t \geq 0$.	Parabola	Constant acceleration	$\frac{1}{2}t^2$	$\frac{1}{s^3}$

Introduction: Evaluating Steady-State Errors



Ex. The concept of e_{ss} for a step and a ramp input.

Step input.

Output 1 has zero steady-state error;
Output 2 has a finite $e_{ss} = e_2(\infty)$.

Ramp input.

Output 1, has zero steady-state error;
Output 2, has a finite $e_{ss} = e_2(\infty)$;
If the output's slope is different from that of the input, then Output 3 results.
Here the steady-state error is infinite.

Errors are measured vertically between the Input and Output after the transients have died down.

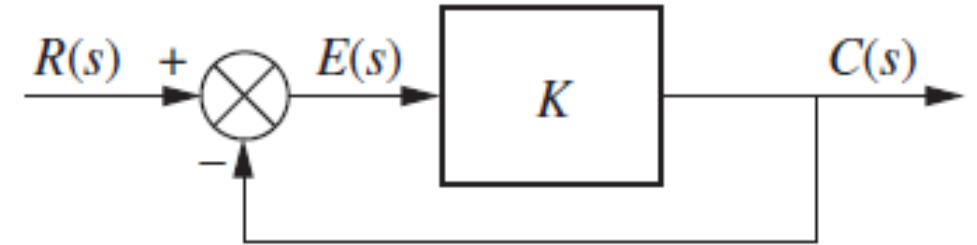
Introduction: Sources of Steady-State Errors

- Many steady-state errors in control systems arise from **nonlinear** sources, however, this topic is beyond the scope of a text on linear control systems.
- The steady-state errors we study here are errors that arise from the **configuration** of the system itself and the type of applied **input**.

Introduction: Sources of Steady-State Errors

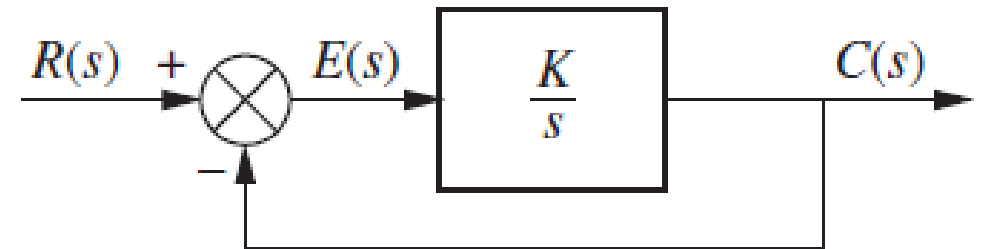
Ex. Step input. Consider the following systems. $E(s) = R(s) - C(s)$ is the error.

In the steady state, if $c(t)$ equals $r(t)$, $e(t)$ will be zero. But with a **pure gain**, K , the error, $e(t)$, cannot be zero if $c(t)$ is to be finite and nonzero.



$$e_{ss} = \frac{1}{K} c_{ss} \quad \text{the steady-state value of the output}$$

If the forward-path gain is replaced by an **integrator**, there will be zero error in the steady state for a step input.



- As $c(t)$ increases, $e(t)$ will decrease, since $e(t) = r(t) - c(t)$.
- This decrease will continue until there is zero error, but there will still be a value for $c(t)$ since an integrator can have a constant output without any input.

Steady-State Error for Unity Feedback Systems: In Terms of $T(s)$

To find $E(s)$, we write

$$E(s) = R(s) - C(s)$$

and we have

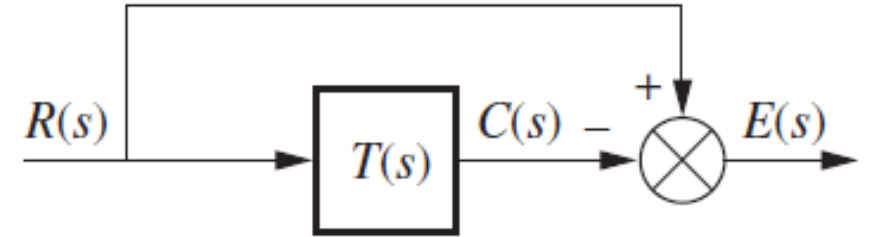
$$C(s) = R(s)T(s)$$

By substituting, simplifying, and solving for $E(s)$

$$E(s) = R(s)[1 - T(s)]$$

Here, we are interested in the final value of the error, $e(\infty)$. Applying the final value theorem, we obtain

$$\begin{aligned} e(\infty) &= \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s) \\ &= \lim_{s \rightarrow 0} sR(s)[1 - T(s)] \end{aligned}$$



Steady-State Error for Unity Feedback Systems: In Terms of $T(s)$

Ex. Find the steady-state error for the system with closed-loop transfer function as follows, if the input is a unit step.

$$T(s) = \frac{5}{s^2 + 7s + 10}$$

Solution:

From the problem statement, $R(s) = \frac{1}{s}$ and $T(s) = \frac{5}{s^2 + 7s + 10}$

$$E(s) = R(s)[1 - T(s)] \quad \rightarrow \quad E(s) = \frac{s^2 + 7s + 5}{s(s^2 + 7s + 10)}$$

Since $T(s)$ is stable and, subsequently, $E(s)$ does not have right-half-plane poles or $j\omega$ poles other than at the origin, we can apply the final value theorem which gives $e(\infty) = \frac{1}{2}$.

Steady-State Error for Unity Feedback Systems: In Terms of $G(s)$

- Many times, we have the system configured as a unity feedback system with a forward transfer function, $G(s)$.
- We find **more insight** for analysis and design by expressing the steady-state error in terms of $G(s)$ rather than $T(s)$.
- Writing $E(s)$, we obtain:

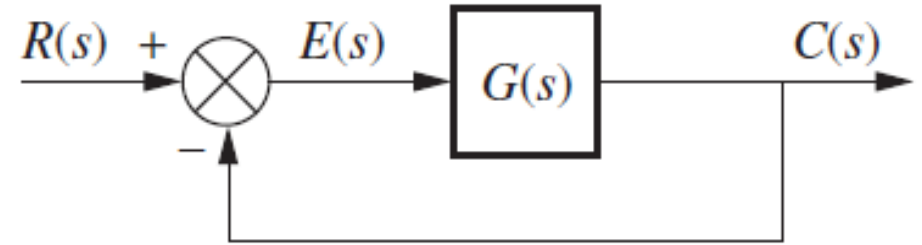
$$E(s) = R(s) - C(s)$$

and we have

$$C(s) = E(s)G(s)$$

substituting and solving for $E(s)$ yields

$$E(s) = \frac{R(s)}{1 + G(s)}$$



Applying the final value theorem, we obtain

$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}$$

Steady-State Error for Unity Feedback Systems: In Terms of $G(s)$

- We now find out the relationships that exist between the open-loop system, $G(s)$, and the nature of the steady-state error, $e(\infty)$ for step, ramp, and parabolic inputs.
- The ability of a system to follow these inputs is indicative of the goodness of the system, because actual inputs may frequently be considered combinations of such inputs.
- **System Type.** Given the open-loop transfer function $G(s)$ in the following form:

$$G(s) \equiv \frac{(s + z_1)(s + z_2) \cdots}{s^n (s + p_1)(s + p_2) \cdots}$$

we define system type to be the value of n in the denominator or, equivalently, the **number of pure integrations** in the forward path.

We will see that the steady-state errors are dependent upon this number of integrations in the forward path.

Steady-State Error for Unity Feedback Systems: In Terms of $G(s)$

Step Input. With $R(s) = 1/s$, we find

$$e(\infty) = e_{\text{step}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s)}{1 + G(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)}$$

In order to have zero steady-state error, we should have

$$\lim_{s \rightarrow 0} G(s) = \infty$$

that means

$$n \geq 1$$

In summary, for a step input to a unity feedback system, the steady-state error will be zero if there is at least one pure integration in the forward path.

If there are no integrations, then there will be a nonzero finite error.

Steady-State Error for Unity Feedback Systems: In Terms of $G(s)$

Ramp Input. With $R(s) = 1/s^2$, we obtain

$$e(\infty) = e_{\text{ramp}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)}$$

To have zero steady-state error for a ramp input, we must have

$$\lim_{s \rightarrow 0} sG(s) = \infty$$

that means

$$n \geq 2$$

In other words, there must be at least two integrations in the forward path.

If there are only **one** integrations in the forward path, the e_{ss} is a **constant** value.

If there are **no** integrations in the forward path, the e_{ss} is **infinite**.

Steady-State Error for Unity Feedback Systems: In Terms of $G(s)$

Parabolic Input. With $R(s) = 1/s^3$, we obtain

$$e(\infty) = e_{\text{parabola}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^3)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)} = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)}$$

To have zero steady-state error for a parabolic input, we must have

$$\lim_{s \rightarrow 0} s^2 G(s) = \infty$$

that means

$$n \geq 3$$

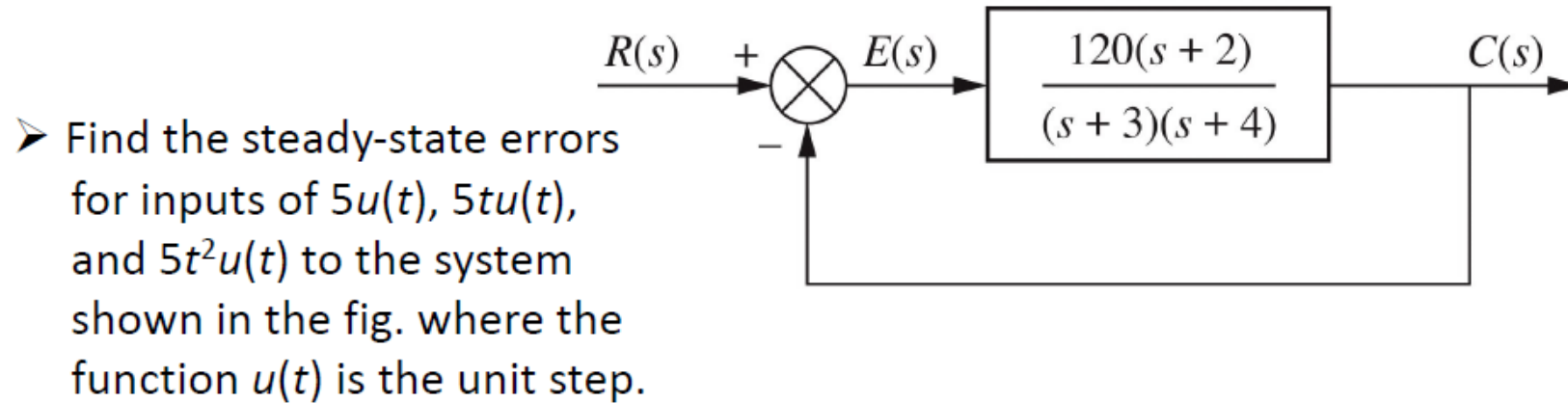
In other words, there must be at least three integrations in the forward path.

If there are only **two** integrations in the forward path, the e_{ss} is a **constant** value.

If there is only **one or less** integration in the forward path, the e_{ss} is **infinite**.

Steady-State Error for Unity Feedback Systems: In Terms of $G(s)$

Ex.



➤ For the input $5u(t)$, whose Laplace transform is $5/s$, the steady-state error will be

$$e(\infty) = e_{step}(\infty) = \frac{5}{1 + \lim_{s \rightarrow 0} G(s)} = \frac{5}{1 + 20} = \frac{5}{21}$$

➤ For the input $5tu(t)$, whose Laplace transform is $5/s^2$, the steady-state error will be

$$e(\infty) = e_{ramp}(\infty) = \frac{5}{\lim_{s \rightarrow 0} sG(s)} = \frac{5}{0} = \infty$$

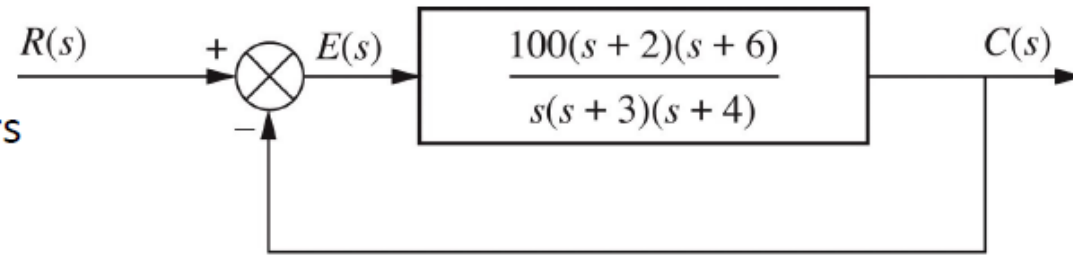
➤ For the input $5t^2u(t)$, whose Laplace transform is $10/s^3$, the steady-state error will be

$$e(\infty) = e_{parabola}(\infty) = \frac{10}{\lim_{s \rightarrow 0} s^2 G(s)} = \frac{10}{0} = \infty$$

Steady-State Error for Unity Feedback Systems: In Terms of $G(s)$

Ex.

- Find the steady-state errors for inputs of $5u(t)$, $5tu(t)$, and $5t^2u(t)$ to the system shown in the fig. where the function $u(t)$ is the unit step, where the function $u(t)$ is the unit step.



- For the input $5u(t)$, whose Laplace transform is $5/s$, the steady-state error will be

$$e(\infty) = e_{step}(\infty) = \frac{5}{1 + \lim_{s \rightarrow 0} G(s)} = \frac{5}{\infty} = 0$$

- For the input $5tu(t)$, whose Laplace transform is $5/s^2$, the steady-state error will be

$$e(\infty) = e_{ramp}(\infty) = \frac{5}{\lim_{s \rightarrow 0} sG(s)} = \frac{5}{100} = \frac{1}{20}$$

- For the input $5t^2u(t)$, whose Laplace transform is $10/s^3$, the steady-state error will be

$$e(\infty) = e_{parabola}(\infty) = \frac{10}{\lim_{s \rightarrow 0} s^2 G(s)} = \frac{10}{0} = \infty$$

Static Error Constants and System Type

In the relationships achieved for steady-state errors, the three terms in the denominator that are taken to the limit determine the steady-state error.

We call these limits **static error constants**. Individually, their names are

- Position constant, K_p , where

$$K_p = \lim_{s \rightarrow 0} G(s)$$

- Velocity constant, K_v , where

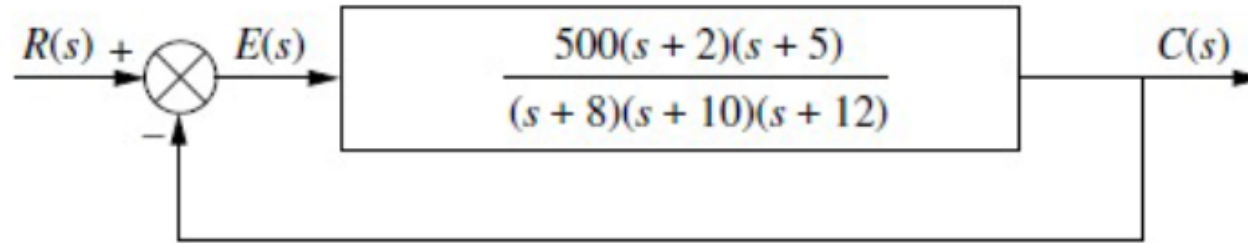
$$K_v = \lim_{s \rightarrow 0} sG(s)$$

- Acceleration constant, K_a , where

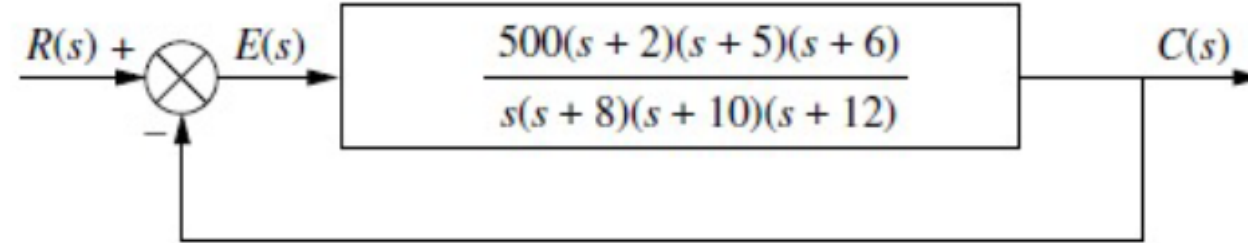
$$K_a = \lim_{s \rightarrow 0} s^2 G(s)$$

Static Error Constants and System Type

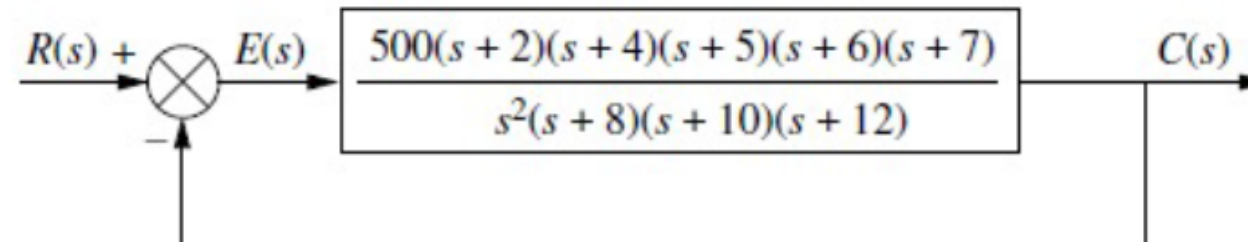
Ex.



(a)



(b)



(c)

Evaluate the static error constants and find the expected error for the standard step, ramp, and parabolic inputs.

Static Error Constants and System Type

Ex (Cont.). Sln.

For the system (a)

$$K_p = \lim_{s \rightarrow 0} G(s) = \frac{500 \times 2 \times 5}{8 \times 10 \times 12} = 5.208$$

$$K_v = \lim_{s \rightarrow 0} sG(s) = 0$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = 0$$

$$e(\infty) = \frac{1}{1 + K_p} = 0.161$$

$$e(\infty) = \frac{1}{K_v} = \infty$$

$$e(\infty) = \frac{1}{K_a} = \infty$$

For the system (b)

$$K_p = \lim_{s \rightarrow 0} G(s) = \infty$$

$$K_v = \lim_{s \rightarrow 0} sG(s) = \frac{500 \times 2 \times 5 \times 6}{8 \times 10 \times 12} = 31.25$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = 0$$

$$e(\infty) = \frac{1}{1 + K_p} = 0$$

$$e(\infty) = \frac{1}{K_v} = \frac{1}{31.25} = 0.032$$

$$e(\infty) = \frac{1}{K_a} = \infty$$

For the system (c)

$$K_p = \lim_{s \rightarrow 0} G(s) = \infty$$

$$K_v = \lim_{s \rightarrow 0} sG(s) = \infty$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = \frac{500 \times 2 \times 4 \times 5 \times 6 \times 7}{8 \times 10 \times 12} = 875$$

$$e(\infty) = \frac{1}{1 + K_p} = 0$$

$$e(\infty) = \frac{1}{K_v} = 0$$

$$e(\infty) = \frac{1}{K_a} = \frac{1}{875} = 1.14 \times 10^{-3}$$

Static Error Constants and System Type

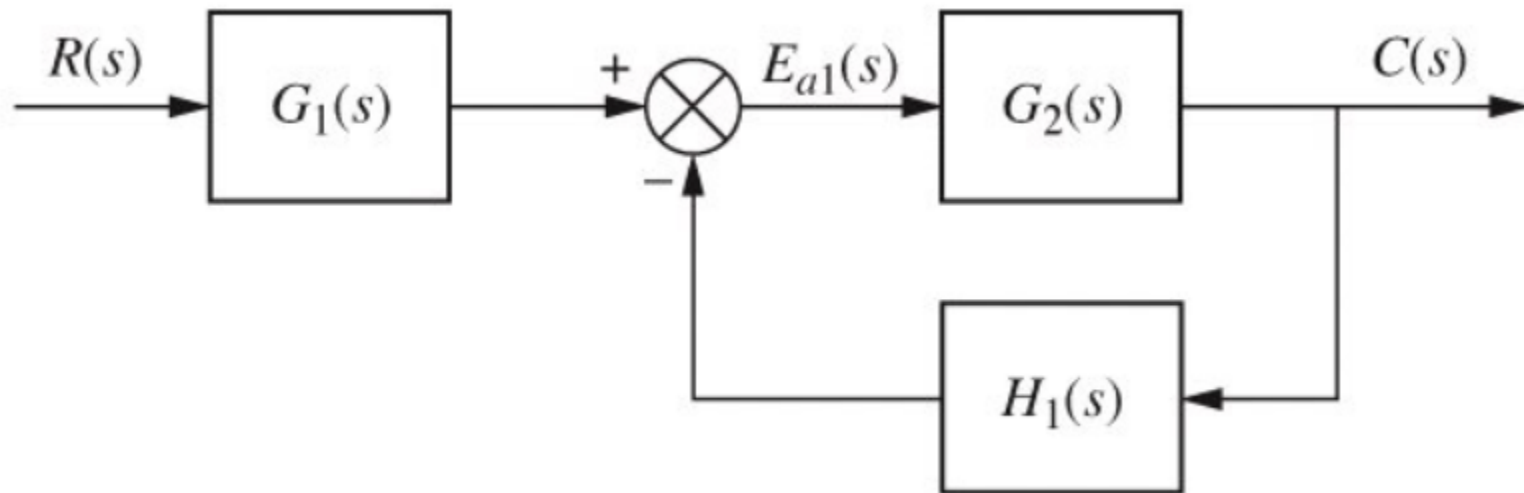
Summary. Relationships between input, system type, static error constants, and steady-state errors

- The values of the static error constants, again, depend upon the form of $G(s)$, especially the number of pure integrations in the forward path.

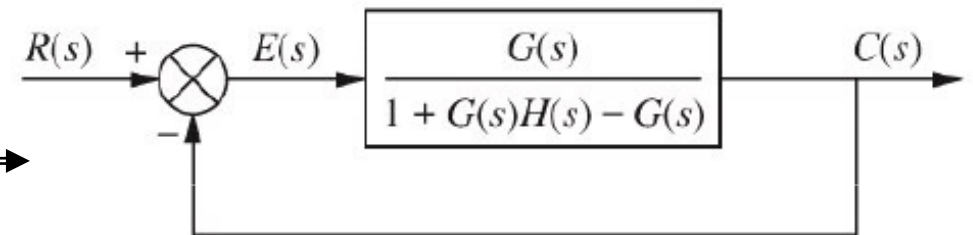
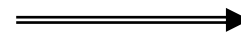
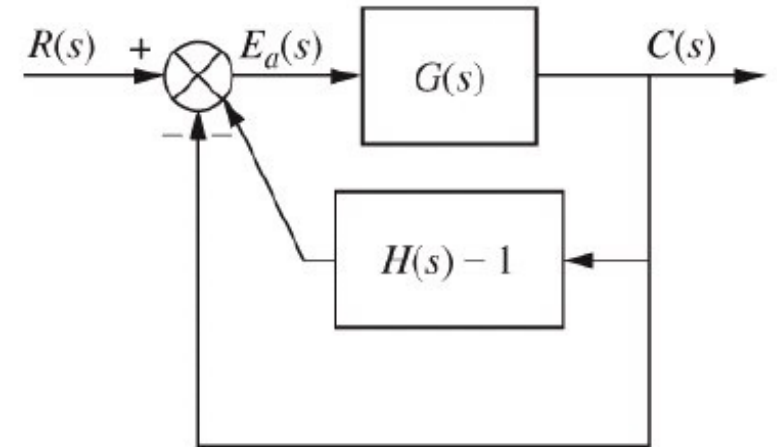
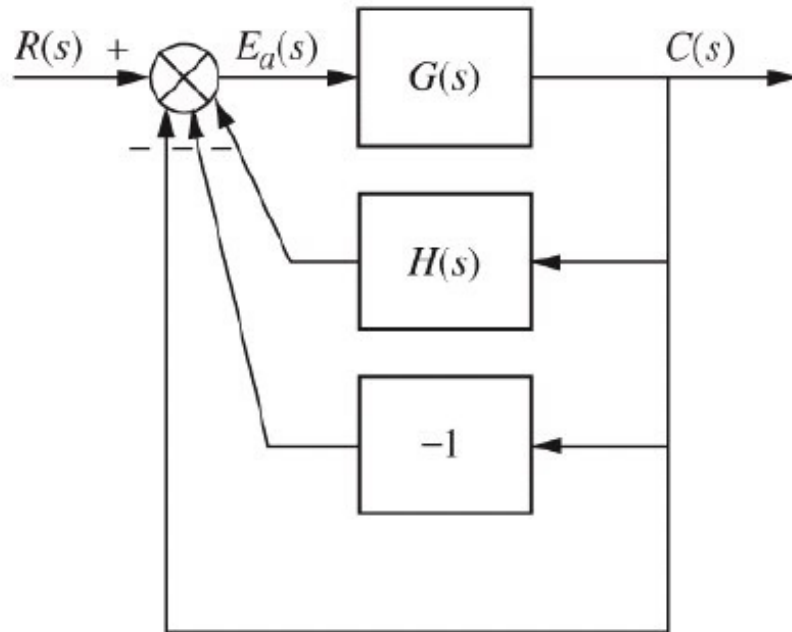
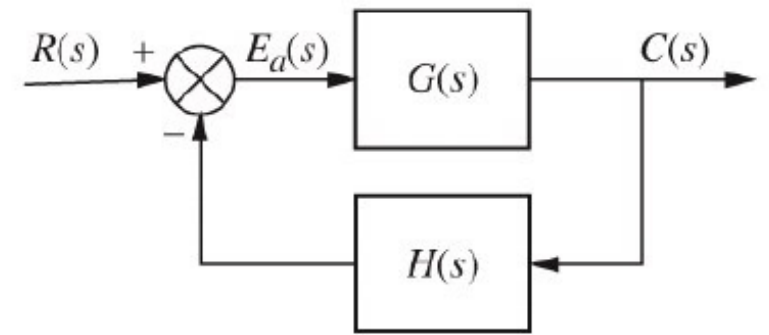
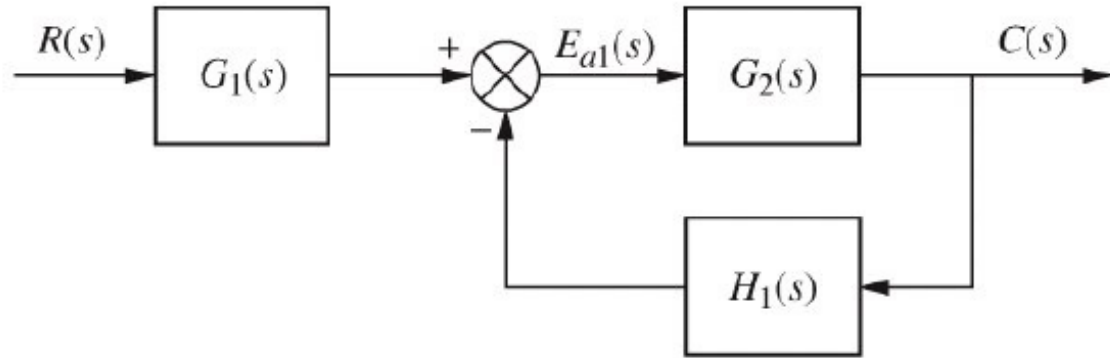
Input	Steady-state error formula	Type 0		Type 1		Type 2	
		Static error constant	Error	Static error constant	Error	Static error constant	Error
Step, $u(t)$	$\frac{1}{1 + K_p}$	$K_p = \text{Constant}$	$\frac{1}{1 + K_p}$	$K_p = \infty$	0	$K_p = \infty$	0
Ramp, $tu(t)$	$\frac{1}{K_v}$	$K_v = 0$	∞	$K_v = \text{Constant}$	$\frac{1}{K_v}$	$K_v = \infty$	0
Parabola, $\frac{1}{2}t^2u(t)$	$\frac{1}{K_a}$	$K_a = 0$	∞	$K_a = 0$	∞	$K_a = \text{Constant}$	$\frac{1}{K_a}$

Steady-State Error for Non-unity Feedback Systems

Control systems often do not have unity feedback because of the compensation used to improve performance or because of the physical model for the system. The feedback path can be a pure gain other than unity or have some dynamic representation.

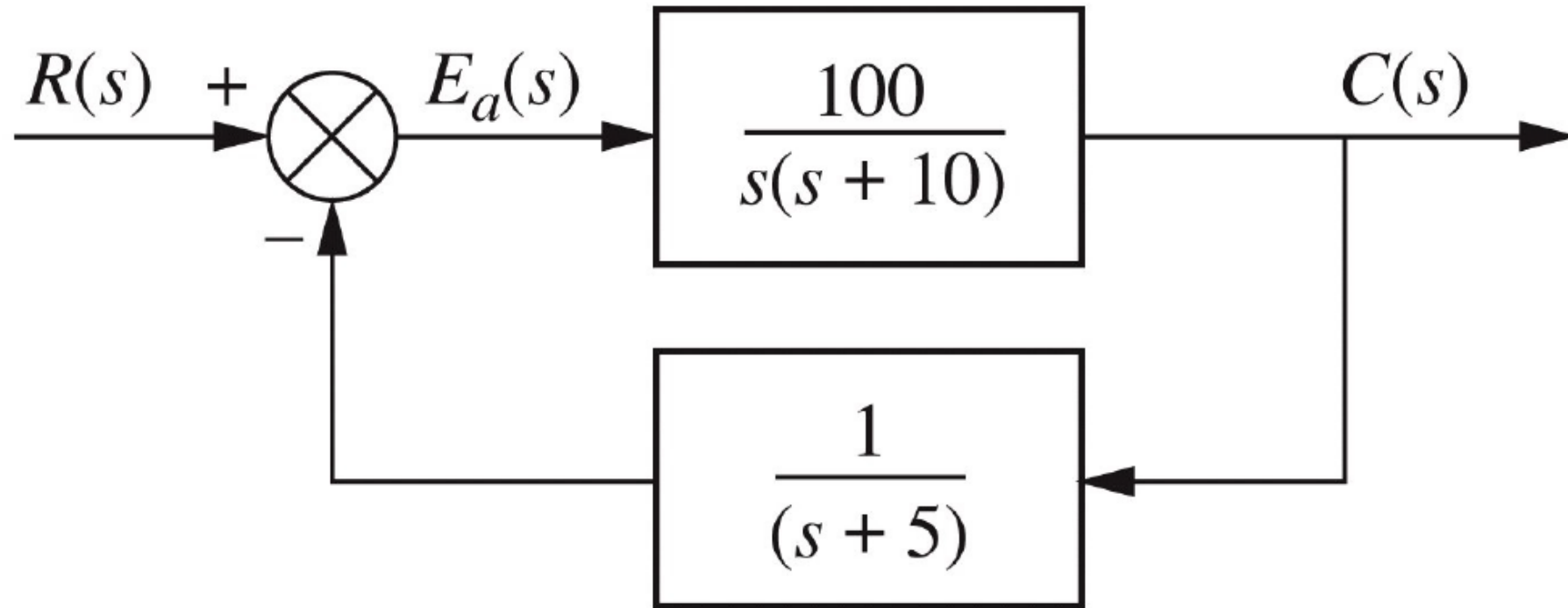


Steady-State Error for Non-unity Feedback Systems



Steady-State Error for Non-unity Feedback Systems

Problem: For the system shown below, find the system type, the appropriate error constant associated with the system type, and the steady-state error for a unit step input. Assume input and output units are the same.



Steady-State Error for Non-unity Feedback Systems

After determining that the system is indeed stable, one may impulsively declare the system to be Type 1. This may not be the case, since there is a nonunity feedback element, and the plant's actuating signal is not the difference between the input and the output.

The first step in solving the problem is to convert the system into an equivalent unity feedback system. Using the equivalent forward transfer function along with

$$G(s) = \frac{100}{s(s+10)} \quad H(s) = \frac{1}{(s+5)}$$
$$G_e(s) = \frac{G(s)}{1 + G(s)H(s) - G(s)} = \frac{100(s+5)}{s^3 + 15s^2 - 50s - 400}$$

Thus, the system is Type 0, since there are no pure integrations. The appropriate static error constant is then K_p , whose value is

$$K_p = \lim_{s \rightarrow 0} G_e(s) = \frac{100 \times 5}{-400} = -\frac{5}{4}$$

Steady-State Error for Non-unity Feedback Systems

The steady-state error is

$$e(\infty) = \frac{1}{1 + K_p} = \frac{1}{1 - (5/4)} = -4$$

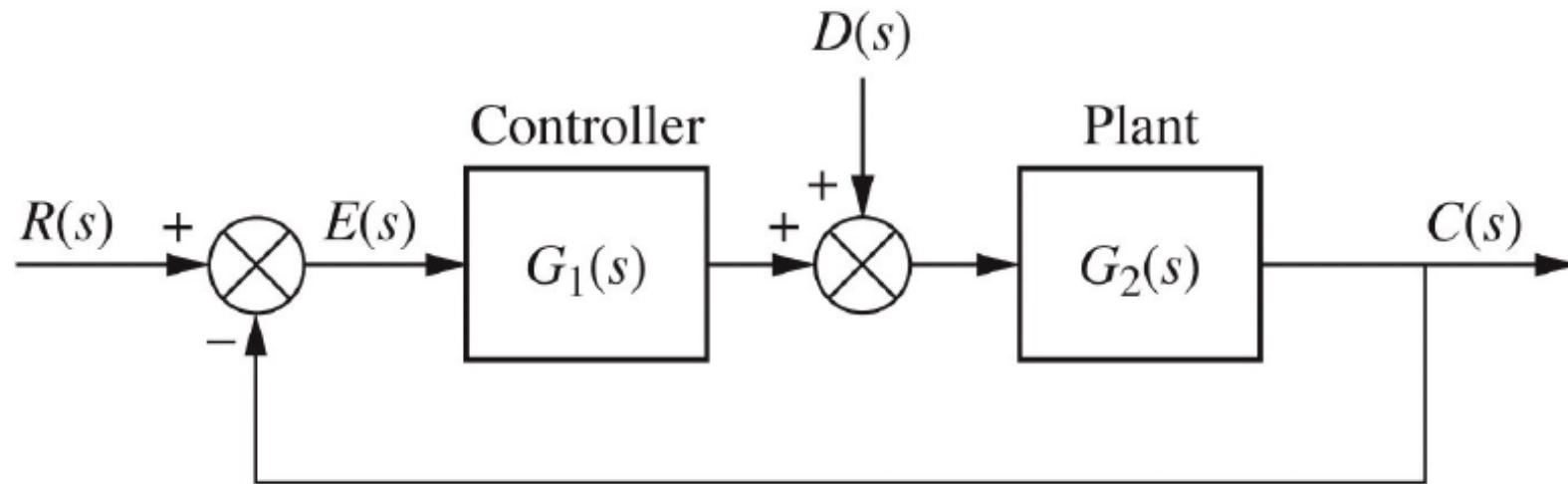
The negative value for steady-state error implies that the output step is larger than the input step.

Alternatively, the same problem can be solved for e_{ss} using the formula with closed-loop transfer function,

$$e(\infty) = \lim_{s \rightarrow \infty} sR(s)[1 - T(s)]$$

Steady-State Errors due to Disturbances

- Feedback control systems are used to compensate for disturbances or unwanted inputs that enter a system.
- The advantage of using feedback is that regardless of these disturbances, the system can be designed to follow the input with small or zero error.
- The figure shows a feedback control system with a disturbance, $D(s)$, between the controller and the plant.



- We now re-derive the expression for steady-state error with the disturbance included. The transform of the output is given by,

Steady-State Errors due to Disturbances

The transform of the output is given by

$$C(s) = E(s)G_1(s)G_2(s) + D(s)G_2(s)$$

and

$$C(s) = R(s) - E(s)$$

Substituting the second equation into the first,

$$E(s) = \frac{1}{1 + G_1(s)G_2(s)} R(s) - \frac{G_2(s)}{1 + G_1(s)G_2(s)} D(s)$$

As seen, the error expression has two terms, which can be considered as two transfer functions; the first one relating $E(s)$ to $R(s)$ and the second to $D(s)$.

In order to find the steady-state error, we can use the Final Value Theorem (FVT) as we did before. FVT is defined as

$$e_{ss} = e(\infty) = \lim_{s \rightarrow 0} sE(s)$$

Steady-State Errors due to Disturbances

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{s}{1 + G_1(s)G_2(s)} R(s) - \lim_{s \rightarrow 0} \frac{sG_2(s)}{1 + G_1(s)G_2(s)} D(s) \\ &= e_R(\infty) + e_D(\infty) \end{aligned}$$

As seen, the rhs consists of two terms: $e_R(\infty)$ and $e_D(\infty)$

$$e_R(\infty) = \lim_{s \rightarrow 0} \frac{s}{1 + G_1(s)G_2(s)} R(s)$$

and

$$e_D(\infty) = -\lim_{s \rightarrow 0} \frac{sG_2(s)}{1 + G_1(s)G_2(s)} D(s)$$

The first one is the error component due to the input $R(s)$, which is already obtained. And the second one is the error due to the disturbance input $D(s)$.

Now let us evaluate this second error component, $e_D(\infty)$ for unit step input,

$$D(s) = \frac{1}{s}$$

Steady-State Errors due to Disturbances

$$\begin{aligned} e_{ss} &= \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} \frac{s}{1 + G_1(s)G_2(s)} R(s) - \lim_{s \rightarrow 0} \frac{sG_2(s)}{1 + G_1(s)G_2(s)} D(s) \\ &= e_R(\infty) + e_D(\infty) \end{aligned}$$

As seen, the rhs consists of two terms: $e_R(\infty)$ and $e_D(\infty)$

$$e_R(\infty) = \lim_{s \rightarrow 0} \frac{s}{1 + G_1(s)G_2(s)} R(s)$$

and

$$e_D(\infty) = -\lim_{s \rightarrow 0} \frac{sG_2(s)}{1 + G_1(s)G_2(s)} D(s)$$

The first one is the error component due to the input $R(s)$, which is already obtained. And the second one is the error due to the disturbance input $D(s)$.

Now let us evaluate this second error component, $e_D(\infty)$ for unit step input,

$$D(s) = \frac{1}{s}$$

Steady-State Errors due to Disturbances

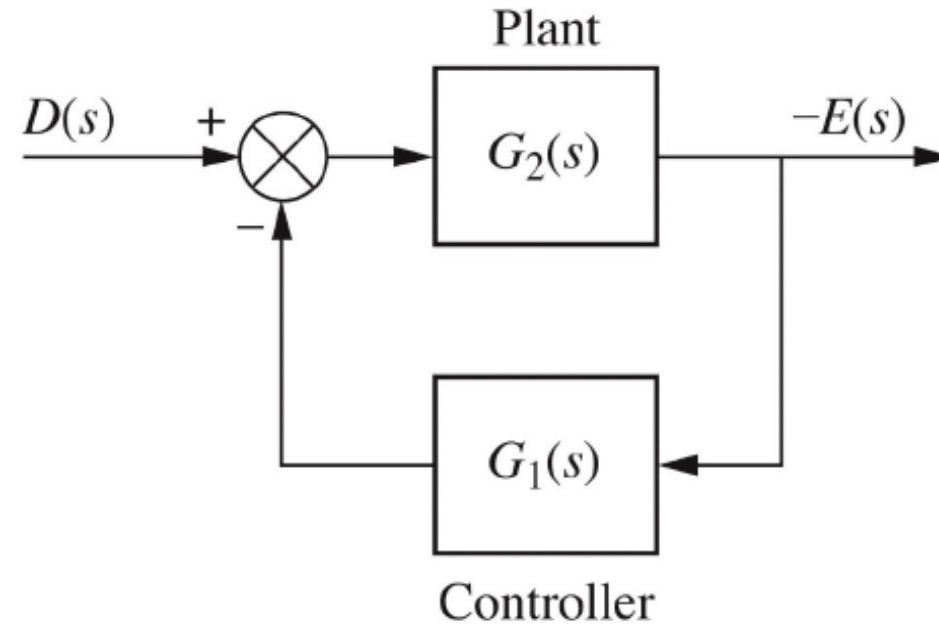
The steady-state error component due to a step disturbance is found to be

$$e_D(\infty) = -\frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)}$$

This equation shows that the steady-state error produced by a step disturbance can be reduced by increasing the dc gain of $G_1(s)$ or decreasing the dc gain of $G_2(s)$.

Steady-State Errors due to Disturbances

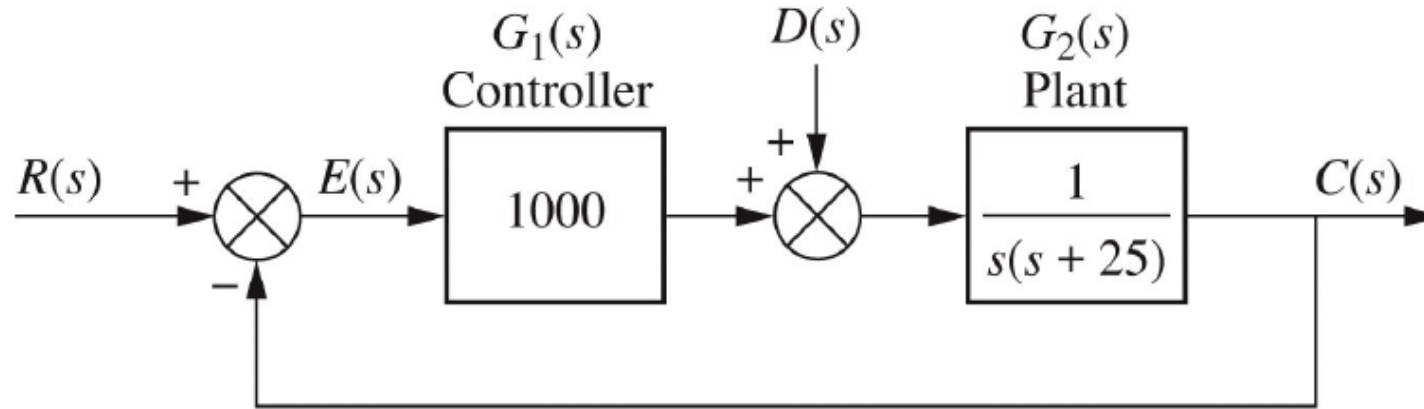
The previous block diagram, where the disturbance input enters the system between the controller and the plant can be converted to the block diagram on the right for $R(s) = 0$. So that we can analyze e_{ss} due to the disturbance input, $D(s)$.



If we want to minimize the steady-state value of $E(s)$, now it is the output in the last figure, we must either increase the dc gain of $G_1(s)$ so that a lower value of $E(s)$ will be fed back to match the steady-state value of $D(s)$, or decrease the dc value of $G_2(s)$, which then yields a smaller value of e_{ss} as predicted by the feedback formula.

Steady-State Errors due to Disturbances

Problem: Find the steady-state error component due to a step disturbance for the system given below.



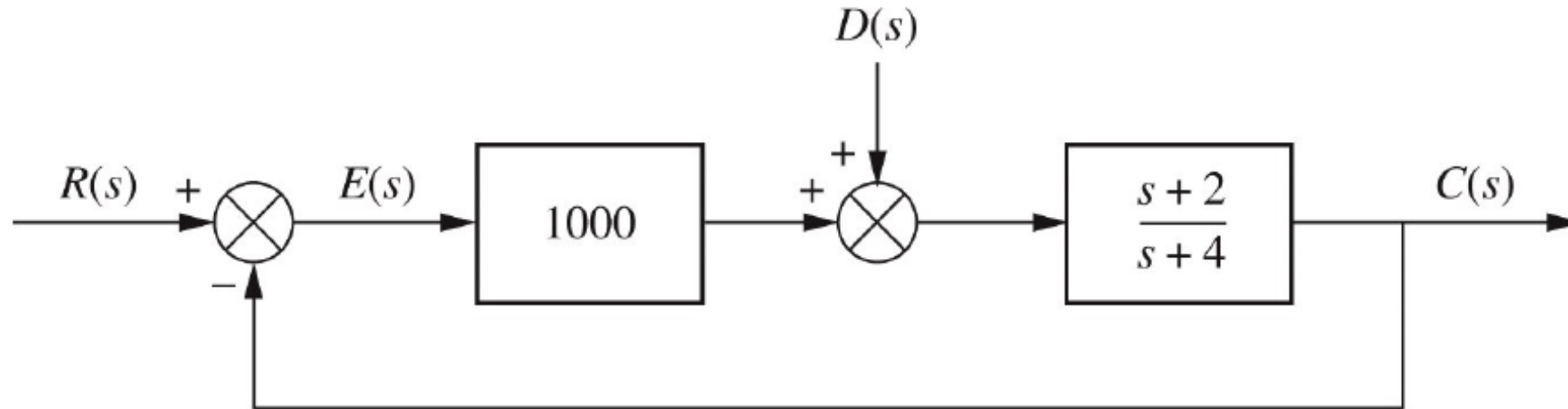
Solution:

$$\begin{aligned} e_D(\infty) &= -\frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)} = -\frac{1}{\lim_{s \rightarrow 0} (s(s+25)) + \lim_{s \rightarrow 0} 1000} \\ &= -\frac{1}{0 + 1000} = -0.001 \end{aligned}$$

- The result shows that the steady-state error produced by the step disturbance is inversely proportional to the dc gain of $G_1(s)$.
- The dc gain of $G_2(s)$ is infinite in this example.

Steady-State Errors due to Disturbances

Problem: Evaluate the steady-state error component due to a step disturbance for the system given below.



Answer: $e_D(\infty) = -9.98 \times 10^{-4}$